

Anti-inflammatory and anti-oxidative effects of cherries on Freund's adjuvant-induced arthritis in rats.



OBJECTIVES: To investigate the anti-inflammatory and anti-oxidative effects of anthocyanins from cherries on Freund's adjuvant-induced arthritis (AIA) in rats. **METHODS:** Arthritis was induced intradermally by injection with 0.1 mL of complete Freund's adjuvant (CFA) into the right hind footpad of male Sprague Dawley (SD) rats. Anthocyanins at 40, 20 and 10 mg/kg (body weight) were administered orally to the treated rats for 28 days after the injection. Tumour necrosis factor-alpha (TNFalpha) in serum and prostaglandin E2 (PGE2) in paws were assayed by radioimmunoassay (RIA), and anti-oxidative effects was assayed by measuring total anti-oxidative capacity (T-AOC), superoxide dismutase (SOD) and malondialdehyde (MDA) in serum. **RESULTS:** Anthocyanins at 40 mg/kg significantly decreased the levels of TNFalpha in serum and PGE2 in paws, simultaneously improving the anti-oxidative status of AIA. We found that at this dosage T-AOC was potentized, the activity of SOD increased and the level of MDA in serum decreased. However, anthocyanins at 20 and 10 mg/kg had less effect on the inflammatory factors and anti-oxidative capacity of AIA. **CONCLUSIONS:** Anthocyanins have potential anti-inflammatory and anti-oxidative effects on AIA.